

ATYRAU, Kazakhstan

WMO: 357000

Lat: 47.1161N

Lon: 51.8838E

Elev: -22

StdP: 101.59

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.2	-18.8	-25.7	0.4	-21.6	-22.5	0.5	-18.4	12.8	-1.1	11.7	-3.9	2.8	50	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	37.6	19.8	36.1	19.4	34.6	18.9	22.4	30.0	21.4	30.4	20.6	30.4	3.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.2	14.8	26.1	18.7	13.5	25.3	17.3	12.4	24.8	66.0	29.6	62.1	30.1	59.2	30.5	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.2	8.8	-23.7	39.3	5.4	1.5	-27.6	40.4	-30.7	41.3	-33.7	42.1	-37.6	43.2		
(5)			-24.0	24.8	5.3	1.2	-27.8	25.7	-30.8	26.4	-33.8	27.1	-37.6	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-6.3	-5.4	2.5	11.7	19.4	24.8	27.3	26.1	18.8	10.3	1.7	-3.6		
	DBStd	13.07	6.02	6.65	4.88	4.98	4.70	4.16	3.30	4.17	4.38	4.84	5.66	5.29		
	HDD10.0	1941	506	432	236	36	1	0	0	0	1	57	252	421		
	HDD18.3	3652	764	666	492	204	43	3	0	2	47	251	501	679		
	CDD10.0	2194	0	0	3	88	293	443	536	500	263	66	2	0		
	CDD18.3	862	0	0	0	7	76	197	277	243	60	2	0	0		
	CDH23.3	10128	0	0	0	59	757	2302	3499	2999	499	13	0	0		
	CDH26.7	5012	0	0	0	11	290	1131	1851	1574	154	1	0	0		
(14) Wind	WSAvg	3.9	4.0	4.3	4.4	4.5	4.1	3.8	3.5	3.3	3.4	3.4	3.6	3.9		
(15) Precipitation	PrecAvg	183	15	12	15	16	27	17	10	10	9	16	19	17		
	PrecMax	340	34	27	47	45	90	46	40	38	41	47	50	48		
	PrecMin	113	1	0	0	0	1	0	0	0	0	0	3	1		
	PrecStd	50	10	8	13	11	23	15	10	12	9	13	11	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	7.7	17.3	28.2	34.8	38.5	39.0	39.3	34.1	25.0	14.8	7.1		
		MCWB	3.1	4.7	10.0	15.3	17.3	19.5	20.1	20.1	18.7	13.7	10.2	4.5		
	2%	DB	2.9	4.6	13.8	25.0	32.2	36.5	37.4	37.5	30.4	22.4	12.4	5.4		
		MCWB	1.9	2.4	8.0	14.0	16.7	19.1	20.0	19.8	16.4	12.9	9.2	3.8		
	5%	DB	1.7	3.1	11.2	22.1	30.0	34.7	36.0	35.8	28.1	20.0	10.6	3.8		
		MCWB	0.9	1.5	6.6	12.7	16.1	18.3	19.7	19.3	15.9	12.2	8.0	2.4		
	10%	DB	0.7	2.0	9.1	19.8	27.6	32.7	34.4	34.0	26.2	17.7	8.8	2.4		
		MCWB	0.0	0.7	5.2	11.8	15.4	17.9	19.4	18.9	15.2	11.0	6.6	1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	5.0	10.6	16.4	19.6	22.3	24.4	24.2	20.6	16.4	12.0	5.5		
		MCDB	4.4	6.9	15.6	24.1	27.1	29.1	30.6	28.9	29.3	21.0	13.4	6.5		
	2%	WB	2.1	2.9	8.6	15.1	18.2	21.1	22.6	22.4	18.8	14.5	9.9	4.1		
		MCDB	2.8	4.4	13.2	23.1	27.2	29.9	30.0	30.1	25.3	19.5	11.4	5.3		
	5%	WB	1.1	1.7	7.0	13.9	17.4	20.2	21.6	21.2	17.6	13.3	8.2	2.6		
		MCDB	1.7	2.7	10.6	20.6	26.7	29.9	30.4	31.1	24.5	18.1	10.2	3.5		
	10%	WB	0.1	0.8	5.6	12.5	16.5	19.5	20.8	20.3	16.5	12.0	6.5	1.5		
		MCDB	0.7	1.8	8.6	18.4	25.4	29.2	30.3	30.8	23.8	16.9	8.6	2.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	7.0	8.2	9.6	10.0	10.6	10.8	11.4	10.7	9.1	6.1	4.8		
		MCDBR	4.7	6.4	10.9	12.0	12.1	12.6	12.4	13.0	12.7	11.7	7.7	4.7		
		MCWBR	4.2	4.8	6.3	5.0	4.6	4.4	4.7	5.1	5.4	5.6	4.9	3.9		
	5% WB	MCDBR	4.4	5.7	9.9	10.8	10.5	10.8	10.8	11.3	10.3	10.1	6.8	4.2		
		MCWBR	4.1	4.5	6.1	4.9	4.5	4.7	5.1	5.5	5.7	6.1	4.9	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.358	0.376	0.434	0.496	0.480	0.455	0.455	0.460	0.417	0.398	0.372	0.370		
	taud		2.052	2.006	2.006	1.920	2.030	2.163	2.184	2.139	2.203	2.183	2.166	2.076		
	Ebn at Noon		670	751	768	759	792	816	809	785	784	726	658	602		
	Edh at Noon		98	126	148	178	165	145	141	141	121	106	88	87		
(44) All-Sky Solar Radiation	RadAvg		0.96	1.80	3.28	4.88	6.20	6.88	6.62	5.84	4.43	2.69	1.40	0.82		
	RadStd		0.17	0.32	0.40	0.37	0.45	0.49	0.32	0.27	0.27	0.24	0.18	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+1.33	N/A	-0.73	N/A	N/A	+231	+155
(47) Regional (2 neighbors)		N/A	N/A	N/A	+1.33	N/A	-0.73	N/A	N/A	+231	+155

Nomenclature: See separate page